

Lecture Notes

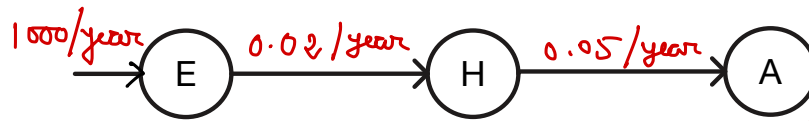
on Leslie Matrix Models, and Long-Term Growth Rate

by Arnaja Mitra

☑ **Example: Population of Locusts:** Consider a population of locusts. Each year, 2% of the **eggs** survive to become **hoppers**, 5% of the hoppers survive to become **adults**, and the adults die after they reproduce (they **do not survive** to the next year). The average female adult locust produces 1000 eggs before she dies. Consider here all locust are female.

Question: How does the population of locusts change over the course of 6 years if

1. There are only 50 adults at the initial time (no eggs or hoppers).
2. There are 50 eggs, 100 hoppers, and 50 adults at the initial time.



Let $E(t)$, $H(t)$ and $A(t)$ denote the population of eggs, hoppers and adults at year t .

$$\text{Let } \vec{x}(t) = \begin{pmatrix} E(t) \\ H(t) \\ A(t) \end{pmatrix}$$

$$\left. \begin{aligned} E(t+1) &= 1000 A(t) \\ H(t+1) &= 0.02 E(t) \\ A(t+1) &= 0.05 H(t) \end{aligned} \right\}$$

we will have this system of equation when we will consider leslie matrix, as soon you will see that leslie matrix is a biological reproduction system whereas for transfer matrix redistribution system. Here quantity flows between states.

$$\vec{x}(t+1) = \begin{pmatrix} E(t+1) \\ H(t+1) \\ A(t+1) \end{pmatrix} = \begin{pmatrix} & & \\ & M & \\ & & \end{pmatrix} \vec{x}(t)$$

$$M = \begin{pmatrix} & & \\ E & H & A \\ 0 & 0 & 1000 \\ 0.02 & 0 & 0 \\ 0 & 0.05 & 0 \end{pmatrix} \begin{matrix} \\ E \\ H \\ A \end{matrix}$$

$$\vec{x}(t) = \begin{pmatrix} E(t) \\ H(t) \\ A(t) \end{pmatrix}$$

→ leslie matrix.

$$1) \vec{x}(0) = \begin{pmatrix} 0 \\ 0 \\ 50 \end{pmatrix}.$$

$$\vec{x}(t+1) = M \vec{x}(t)$$

$$\Rightarrow \vec{x}(1) = \begin{pmatrix} 0 & 0 & 1000 \\ 0.02 & 0 & 0 \\ 0 & 0.05 & 0 \end{pmatrix} \vec{x}(0) = \begin{pmatrix} 0 & 0 & 1000 \\ 0.02 & 0 & 0 \\ 0 & 0.05 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 50 \end{pmatrix} = \begin{pmatrix} 50,000 \\ 0 \\ 0 \end{pmatrix}$$

$$\vec{x}(2) = \underbrace{M \vec{x}(1)} = M(M \vec{x}(0)) = M^2 \vec{x}(0)$$

$$= \begin{pmatrix} 0 & 0 & 1000 \\ 0.02 & 0 & 0 \\ 0 & 0.05 & 0 \end{pmatrix} \begin{pmatrix} 50,000 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1,000 \\ 0 \end{pmatrix}$$

$$\vec{x}(3) = \underbrace{M \vec{x}(2)} = M(M^2 \vec{x}(0)) = M^3 \vec{x}(0)$$

$$= \begin{pmatrix} 0 & 0 & 1000 \\ 0.02 & 0 & 0 \\ 0 & 0.05 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1,000 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 50 \end{pmatrix}$$

Find the values of $\vec{x}(4)$, $\vec{x}(5)$ and $\vec{x}(6)$ at home.

$$2) \vec{x}(0) = \begin{pmatrix} 50 \\ 100 \\ 50 \end{pmatrix}$$

$$\vec{x}(1) = M \vec{x}(0) = \begin{pmatrix} 0 & 0 & 1000 \\ 0.02 & 0 & 0 \\ 0 & 0.05 & 0 \end{pmatrix} \begin{pmatrix} 50 \\ 100 \\ 50 \end{pmatrix}$$

$$= \begin{pmatrix} 50,000 \\ 1 \\ 5 \end{pmatrix}$$

$$\vec{x}(2) = \underbrace{M \vec{x}(1)}_{= M(M \vec{x}(0))} = M^2 \vec{x}(0)$$

$$= \begin{pmatrix} 0 & 0 & 1000 \\ 0.02 & 0 & 0 \\ 0 & 0.05 & 0 \end{pmatrix} \begin{pmatrix} 50,000 \\ 1 \\ 5 \end{pmatrix}$$

$$= \begin{pmatrix} 5,000 \\ 1,000 \\ 0.05 \end{pmatrix}$$

Find values of $\vec{x}(4)$, $\vec{x}(5)$ and $\vec{x}(6)$ at home.

Definition: Leslie Matrix Model

The matrix of a population model that takes into account the births and the survival rates of each class to the next stage (class) is known as a **Leslie matrix model**.

The general form of a Leslie matrix for a population model with n stages is

$$M = \begin{pmatrix} F_1 & F_2 & F_3 & \cdots & F_n \\ S_1 & 0 & 0 & \cdots & 0 \\ 0 & S_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & S_{n-1} & S_n \end{pmatrix}.$$

The system of equations is given by

$$\vec{x}(t+1) = M\vec{x}(t), \quad \text{where } x(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}.$$

Here, F_i ($i \in \{1, 2, \dots, n\}$) is the average number of female offspring born to a female individual during one time step in class i . S_i ($i \in \{1, 2, \dots, n-1\}$) is the fraction of individuals in class i that survive to class $(i+1)$ in the next time step.

For $i = n$, S_n denotes the proportion of individuals in class n that survive to the subsequent time period (if any).

F_i is known as the **fecundity rate**, and S_i is known as the **survival rate**.

Note: If individuals in their final stage (n) die, then $S_n = 0$.

✍ **Example: Cattle Population Model:** Cattle on a large ranch are divided into **calves**, **yearlings**, and **adults**. Data indicate that 70% of the calves **survive** the first year to become yearlings, while 80% of the yearlings mature into adults. In addition, 90% of adults survive a given year, and one adult female produces a single calf each year.

Question: Construct the Leslie matrix, assuming that we census the population after reproduction and count only females.

